BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

ELECTRONICS MATERIALS & INSTRUMENTATION (2024)

Exam Date: Saturday, 7 Nov 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2024	Measurements and Instrumentation	110601	B.Tech 6th Semester Examination, 2024

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 6th Semester Examination, 2024

MEASUREMENTS AND INSTRUMENTATION

Time: 03 Hours | **Code: 110601** | **Full Marks: 70**

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[14]**

1. Which system analyzes brain wave activity?
   i. EOG
   ii. EMG
   iii. ECG
   iv. EEG

2. Which type of sensor is used to measure flow?
   i. Potentiometer
   ii. Venturi tube
   iii. Hall effect sensor
   iv. Thermocouple

3. What does "Six Sigma" focus on?
   i. Enhancing resolution
   ii. Improving precision
   iii. Reducing defects in processes
   iv. Minimizing systematic errors

4. What is the mean of the following data set: 4, 6, 8, 10?
   i. 6
   ii. 2
   iii. 8
   iv. 9

5. Which instrument is used for ultrasonic diagnostic measurement?
   i. Thermocouple
   ii. Oscilloscope
   iii. Ultrasound transducer
   iv. Megger

6. What is an example of a non-invasive Cardiac output measurement diagnosis?
   i. Temperature measurement
   ii. Blood pressure measurement
   iii. ECG
   iv. All of the above

7. What is the function of a transducer in bio-medical instruments?
   i. Convert electrical signals into light
   ii. Measure electric current
   iii. Measure pressure
   iv. Convert biological signals into electrical signals

8. What does a Megger measure?
   i. Current
   ii. Insulation resistance
   iii. Capacitance
   iv. Voltage

9. What does a digital multi-meter measure?
   i. Only current
   ii. Only resistance
   iii. Only voltage
   iv. Voltage, current, and resistance

10. What type of instrument is used to measure inductance?
    i. Ohmmeter
    ii. LCR meter
    iii. Ammeter
    iv. Voltmeter

Q.2 Solve both questions: **[14]**

1. Discuss the use of instrument transformers for current and voltage measurement in power systems.
2. Explain the working principle of Maxwell's bridge with phasor diagram.

Q.3 Solve both questions: **[14]**

1. Explain the Hall effect and its application in Hall sensors for current and voltage measurements.
2. Describe the working principle of a Digital Storage Oscilloscope (DSO).

Q.4 Solve both questions: **[14]**

1. Define and differentiate between true value, accuracy, and precision in measurement systems.
2. Explain the concept of True RMS measurement. Why is it important for accurate AC voltage and current measurements?

Q.5 Solve both questions: **[14]**

1. Explain the Six-Sigma approach to quality control in industrial processes.
2. Describe the working principles of temperature transducers thermocouples and RTD.

Q.6 Solve both questions: **[14]**

1. Explain hysteresis and dead-band in measurement systems. How do they influence the performance of transducers?
2. Explain how bio-electric signals like ECG, EEG, and EMG are generated and measured.

Q.7 Solve both questions: **[14]**

1. Discuss the types of errors that occur in measurements. Explain the difference between systematic and random errors with practical examples.
2. Discuss non-invasive diagnostic measurements used in healthcare.

Q.8 Solve both questions: **[14]**

1. Describe the role of sensitivity in measurement systems. How is sensitivity calculated, and what are the consequences of low or high sensitivity in sensors?
2. Explain the use of Meggers in insulation resistance testing. Why is insulation testing crucial in high-voltage systems?

Q.9 Write short notes on any two of the following: **[14]**

1. a. CAT scan techniques
   b. Cardiovascular measurements
   c. Capacitance Measurement with phasor
   d. LVDT